

Fredson SILVA DE SOUZA AGUIAR

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Objective

Developing high quality research in the field of Control Theory and Applications, with interest in its association with numerical and learning-based methods.

Academic Training

2025 – 2026: Sorbonne Paris Nord (PNGS-M&CS)

Master 2 in Mathematics of Data
University Average Grade (GPA): N/A

2023 – 2025: Escola de Matemática Aplicada (FGV/EMAp)

Master's Degree in Mathematical Modeling
University Average Grade (GPA): 9.25/10.00

2018 – 2022: Escola de Matemática Aplicada (FGV/EMAp)

Bachelor of Science in Applied Mathematics
University Average Grade (GPA): 9.10/10.00

Professional Experience

apr/2026 – aug/2026: Research Internship – FAU MoD, Erlangen

Area: Control Theory, Reinforcement Learning and Optimization Algorithms

jun/2021 – nov/2021: Internship – IBM Brazil

Area: *Natural Language Processing; Repair and Publication of Language Data*

jan/2021 – feb/2021: Summer Job - Bank BOCOM BBM

Area: *Data Science, Credit Analysis and Automation for Financial Data Lake*

aug/2019 – jul/2021: Scientific Initiation – PICME/CNPQ

Area: *Natural Language Processing; Semantic Representations and Ontologies*

Relevant Knowledge

Optimal Control Theory; Applied Mathematics; Data Science; Reinforcement Learning; Neural Network; Data Structures and Algorithms; Mathematical Modelling;

Abstracts and Publications

Fredson S. S. Aguiar, and Maria S. Aronna. A Learning-Based Model Predictive Control Method for Epidemic Mitigation. In X Congresso De Matemática Aplicada, Computacional E Industrial (MACI 2025).

<https://amcaonline.org.ar/maci/index.php/maci2025/maci2025/paper/view/5771>

Fredson S. S. Aguiar, and Maria S. Aronna. A Learning-Based Model Predictive Control Framework with a State Observer for Epidemic Mitigation. In XLIII National Congress of Applied and Computational Mathematics (CNMAC 2024). In Proceeding Series of the Brazilian Society of Computational and Applied Mathematics, v. 11, n. 1, 2025.

<https://proceedings.sbmacc.org.br/sbmacc/article/view/4896>

Hugo Gonalo Oliveira, Fredson Silva de Souza Aguiar, and Alexandre Rademaker. On the Utility of Word Embeddings for Enriching OpenWordNet-PT. In 3rd Conference on Language, Data and Knowledge (LDK 2021). Open Access Series in Informatics

(OASlcs), Volume 93, pp. 21:1-21:13, Schloss Dagstuhl – Leibniz-Zentrum für Informatik (2021) <https://doi.org/10.4230/OASlcs.LDK.2021.21>

Research Projects

Fredson S. S. Aguiar., and Maria S. Aronna. From Uncontrolled to Controlled Predictions: A Learning-Based Model Predictive Control Framework with an State Predictor for Epidemic Mitigation under Dynamics and Parametric Uncertainty. Master's Monograph. <https://hdl.handle.net/10438/37618>

Fredson S. S. Aguiar. A Case Study on Inverse Optimal Control and Model Predictive Control for Autonomous Decision Making Under Timely Distributional Shift. Condor Fellowship. https://fredsonaguiar.github.io/assets/pdf/IOC_MPC.pdf

Extra-curricular Activities

aug/2024 – current: Research Assistant– FGV/EMAp

Description: Biological Optimal Control of Sugarcane Crop's Parasites

aug/2024 – dec/2024: Teaching Assistant– FGV/EMAp

Description: TA in PhD Level Introduction to Functional Analysis

mar/2024 – jun/2024: Teaching Assistant– FGV/EMAp

Description: TA in PhD Level Qualitative Theory of ODEs

aug/2023 – dec/2023: Teaching Assistant – FGV/EMAp

Description: TA in Master Level Multivariate Analysis (Analysis in R^n)

aug/2023 – dec/2023: Teaching Assistant – FGV/EMAp

Description: TA in Undergraduate Level Partial Differential Equations

mar/2023 – jun/2023: Teaching Assistant – FGV/EMAp

Description: TA in Undergraduate Level Complex Calculus

aug/2022 – dec/2022: Teaching Assistant – FGV/EMAp

Description: TA in Undergraduate Level Real Analysis

aug/2020 – dec/2020: Teaching Assistant – FGV/EMAp

Description: TA in Undergraduate Level Discrete Mathematics

feb/2020 – jun/2020: Teaching Assistant – FGV/EMAp

Description: TA in Undergraduate Level Vectorial Calculus

Volunteer Activities

aug/2023 – aug/2025: Coordinator – FGV/SEPEMAp

Description: Graduate Student Seminars for Research Dissemination

jan/2019 – nov/2024: Tutor and Assistant – FGV/CDMC

Description: Talents Selection Social Program as Tutor and Assistant

feb/2020 – nov/2020: Volunteer Teacher - Amplia FGV

Description: Social Project Volunteer as Basic Mathematics Teacher

Relevant Events Participation

sep/2025: 12th Heidelberg Laureate Forum (12HLF)

Description: Young Researcher in the Heidelberg Laureate Forum

dec/2024: Modeling and Numerics in Epidemiology (SaC3)

Description: First Encounter for Summer the School and Workshop

fev/2024: Condor Camp 2024

Description: Bootcamp in Career Acceleration, AI Safety and Governance